UGC & Govt. Approved
Sonargaon University (SU)
সোনারগাঁও ইউনিভার্সিটি (এসইউ)

Imbalance-Aware AutoML for Financial Risk Prediction

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Introduction

Class imbalance in financial datasets causes standard machine learning models to overlook rare but critical events like fraud, default, and churn, leading to substantial business losses.

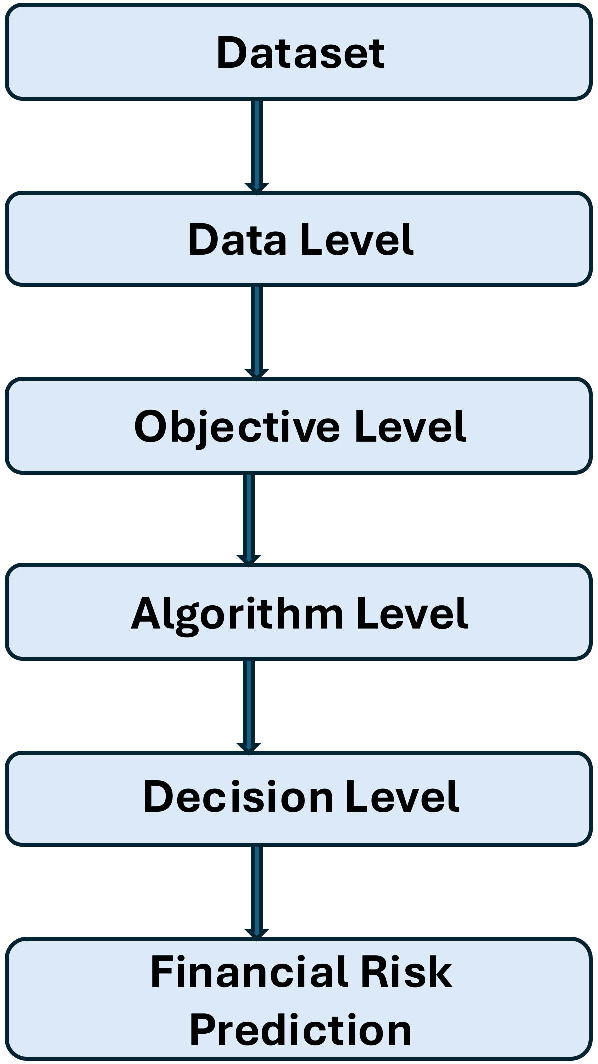
This research develops an AI-powered system to

- Fraud Detection
- Credit Risk Prediction
- Loan Default Prediction
- Customer Churn Prediction
- Financial Decision Making

Problem Statement

Traditional AutoML systems may

- Focus on Accuracy
- Ignore minority classes
- Not automatically handle imbalance
- Use default decision threshold = 0.5



Objectives

The main objectives of this research are

1

A bank wants to see which method is best for detecting Fraud Transactions.

2

To test the model on various real-world problems such as Credit Card Fraud, Loan Default, and Customer Churn.

3

To compare which one gives better predictions using XGBoost-GPU and FLAML on the same financial dataset.

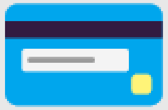
4

To select a strategy that will detect more Fraud or Loan Default and reduce the bank's financial loss.

Motivation

In financial datasets, risky cases are often much fewer than normal cases. This imbalance can cause ML models to overlook critical high-risk customers.

Loan Default



Credit Card Fraud

Insurance Fraud

Research Question

System Performance

Which strategy performs best?

Engine Consistency

Are results consistent across AutoML engines?

Imbalance Effect

How does imbalance severity affect performance?

Business Cost

Which strategy provides high recall and low cost?

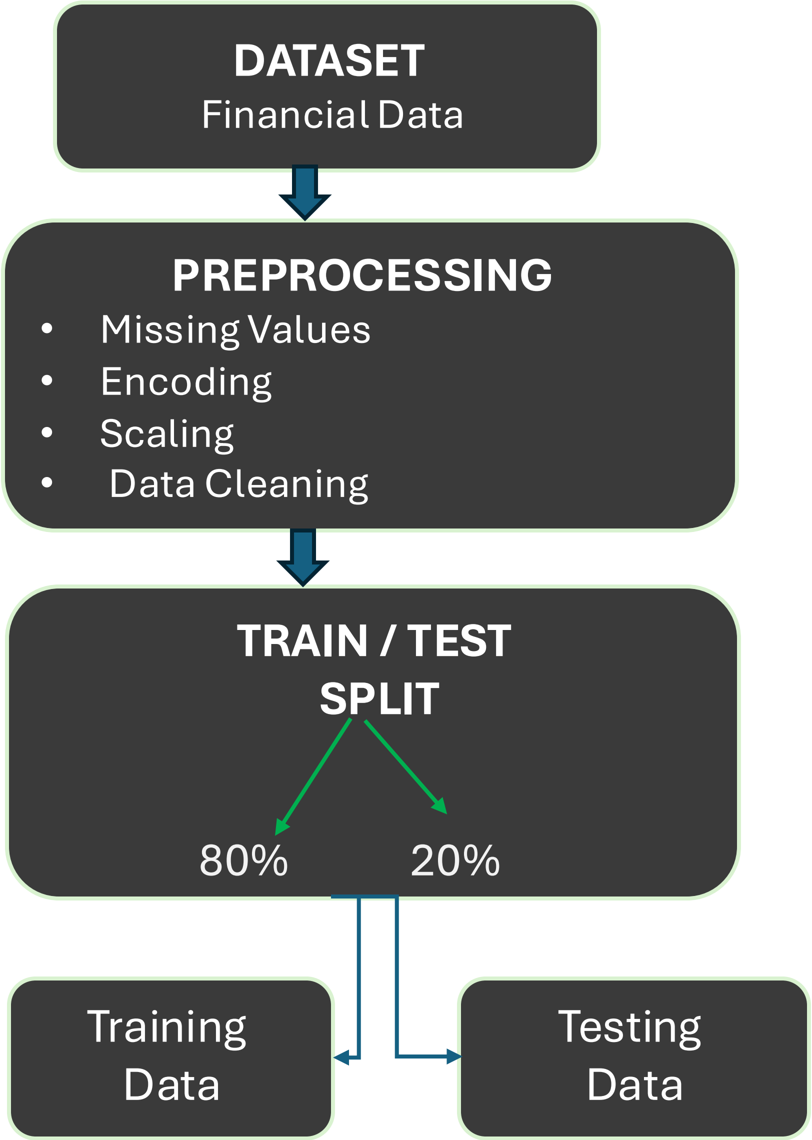
Dataset and Preprocessing

Dataset

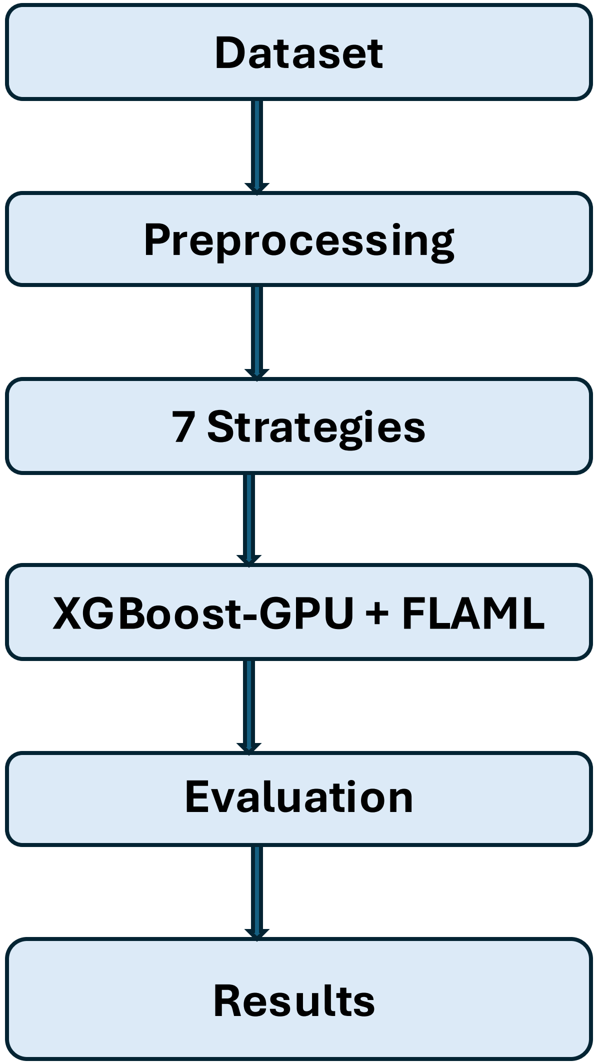
- 8 Public Financial & Business Datasets
- Minority Rate: 0.17% – 44.5%
- Numerical + Categorical Features

Preprocessing

- Missing Value Handling
- Feature Encoding
- Scaling
- Minority Class = 1



Methodology

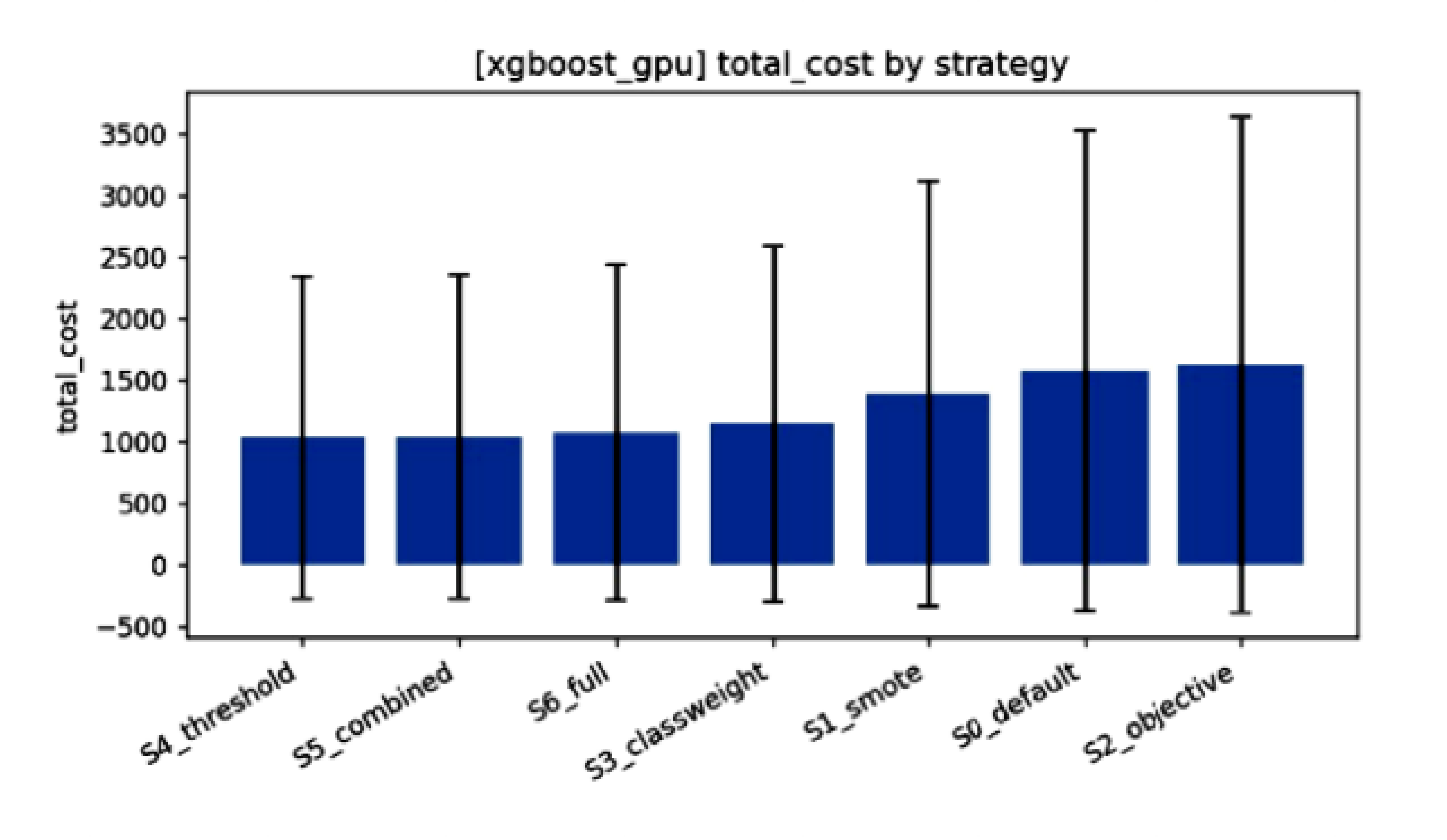


- 7 Imbalance Handling Strategies
- 8 Datasets
- 10 Random Seeds
- 2 AutoML Engines
- Total = 1,120 Experimental Runs

Data Split

- 60% Training
- 20% Validation
- 20% Testing

Experimental Results



Key Findings

- Threshold-based strategies achieved better performance.
- Minority recall increased significantly. Financial cost was reduced.
- PR-AUC remained almost unchanged.

Experimental Results (cont.)

Best Performing Strategies

S3 – Class Weight
S4 – Threshold Tuning
S5 – Combined Strategy

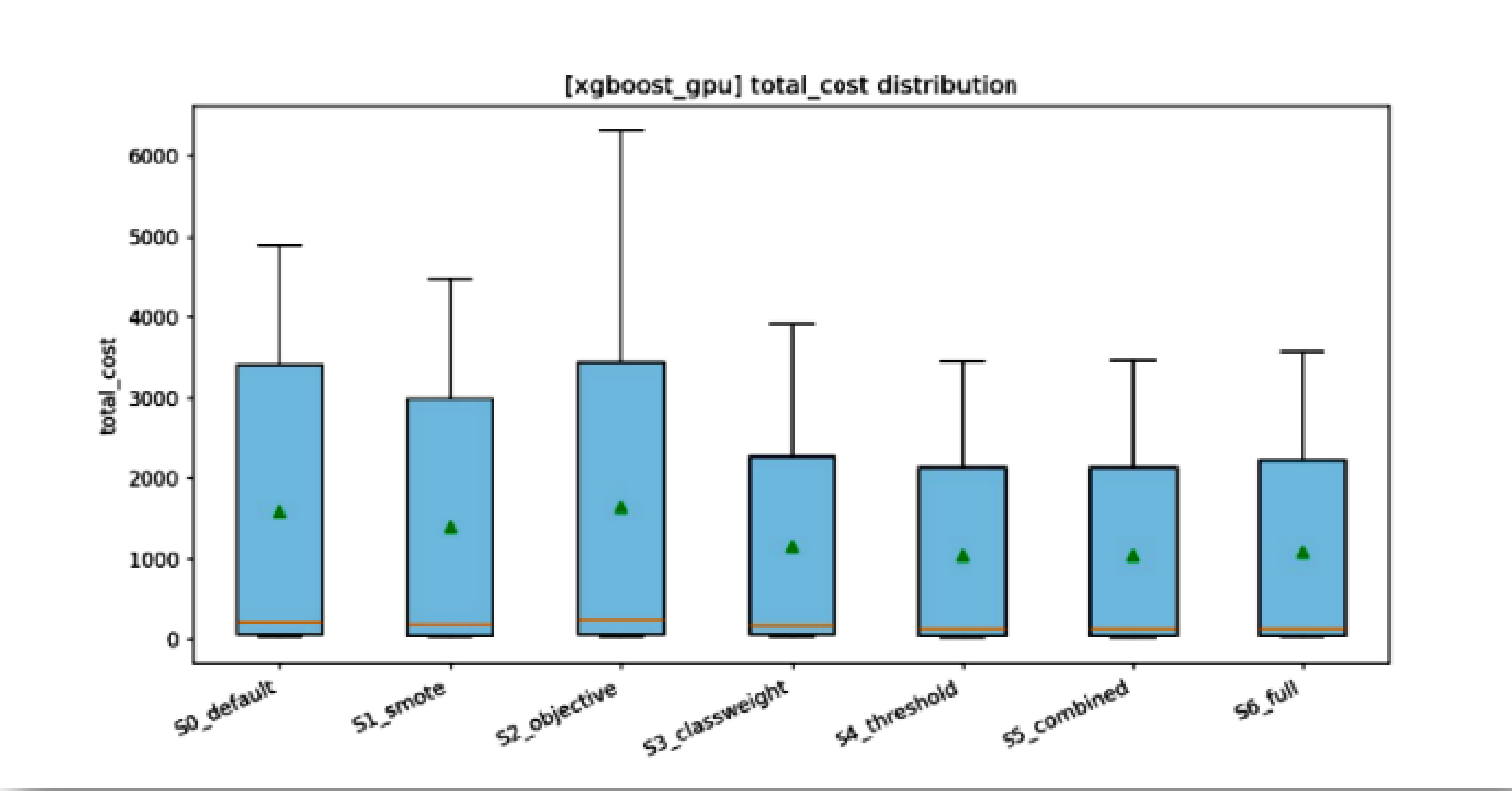


Fig 4.20: Total Cost Distribution by Strategy

Features

Financial Risk Prediction System

Fraud Detection

- Dataset Evidence: Credit Card Fraud Dataset
- Minority Rate: 0.17%

Credit Risk Prediction

- Dataset Evidence: Credit G (German), Australian Credit &Credit Approval Dataset

Default Prediction

- Dataset Evidence: Default Taiwan Dataset
- Default Rate: 22.1%

Customer Churn Prediction

- Dataset Evidence: Churn Dataset
- Churn Rate: 14.1%

Features (cont.)

```
CREDIT CARD FRAUD DATASET ANALYSIS
========================================================================
Total transactions: 284,807
Total features: 29

Target Distribution (0 = Legitimate, 1 = Fraud):
Class
0    284315
1       492
Name: count, dtype: int64

Minority (Fraud) Rate Calculation:
Fraud (1) count: 492
Legitimate (0) count: 284,315
Total transactions: 284,807
Fraud Rate: 0.1727%

Minority Rate (decimal): 0.001727
Minority Rate (percentage): 0.17%
========================================================================
```

+ Code + Markdown

```
CREDIT G (GERMAN) DATASET ANALYSIS
========================================================================
Total rows: 1000
Total features: 20

Target Distribution:
class
good    700
bad     300
Name: count, dtype: int64

Minority Class Calculation:
Minority Label: 'bad'
Minority Count: 300
Total Records: 1000
Minority Rate: 30.00%
========================================================================
```

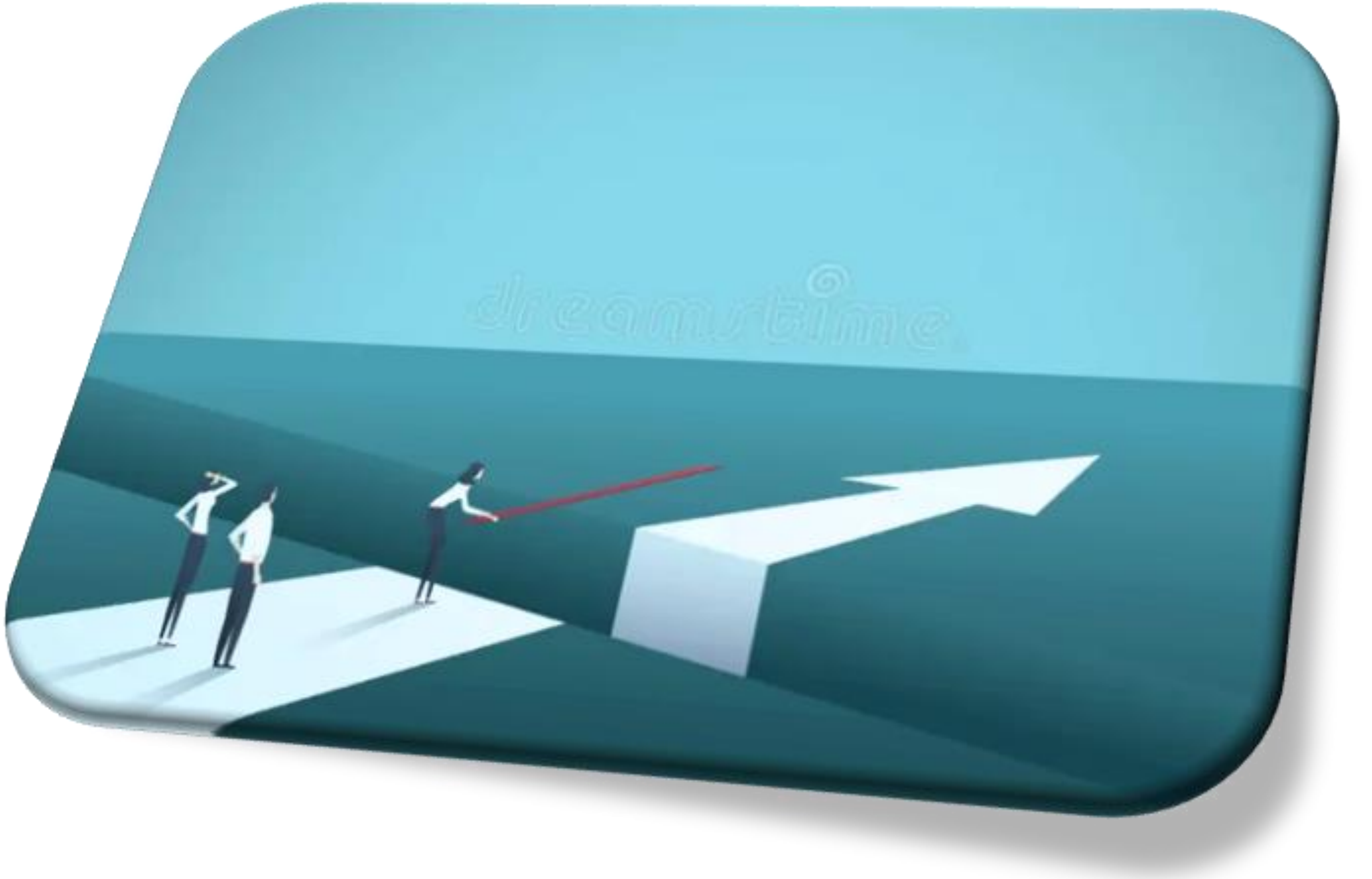
```
DEFAULT TAIWAN DATASET ANALYSIS
========================================================================
Total rows: 30000
Total features: 23

Target Distribution:
y
0    23364
1     6636
Name: count, dtype: int64

Default Rate Calculation:
Default (1) count: 6636
Non-Default (0) count: 23364
Default Rate: 22.12%
```

```
CHURN DATASET ANALYSIS
========================================================================
Total customers: 5000
Total features: 20

Target Distribution:
class
0    4293
1     707
Name: count, dtype: int64

Churn Rate Calculation:
Churn (1) count: 707
No Churn (0) count: 4293
```

Challenges

The major challenges were

- Extreme Class Imbalance
- Minority Class Detection
- Misleading Accuracy
- High Computational Cost
- Selecting the Best Strategy
- Different Business Cost Assumptions



Achievements & Impact

Achievements

- Compared 7 different strategies
- Used 8 real-world datasets
- Completed 1,120 experiments
- Compared 2 AutoML engines
- Improved minority-class recall
- Improvement in Risk Detection
- Financial Cost Reduction
- Reduced financial prediction cost

Impact

- Better Banking Decisions
- Fraud Prevention
- Reduced Default Risk
- Lower Financial Loss

Limitations & Future Works

Limitations:

- Binary Classification Only
- Limited AutoML Engines
- Fixed Cost Ratio
- Limited Search Space

Future Works:

- Dynamic Threshold Tuning
- Deep Learning AutoML
- Multi-Objective Optimization
- Real-Time Financial Prediction
- Explainable AI

Conclusion

Class Imbalance Problem

- Risky cases are usually very few in financial data, so typical ML models often miss important fraud or default cases.

Our Solution

- We tested various imbalance-handling strategies using Imbalance-Aware AutoML.

Key Finding

- Threshold Tuning and Cost-Sensitive Decision Making gave the best and most consistent performance.

Real-World Benefit

- The system can identify more risky cases and help reduce the potential financial cost of wrong decisions.

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5. Kuhn, M., & Johnson, K. (2019). Feature Engineering and Selection: A Practical Approach for Predictive Models. CRC Press.

Dataset

https://www.openml.org/search?type=data&status=active&id=31&sort=runs

https://www.openml.org/search?type=data&status=active&id=29&sort=runs

https://www.openml.org/search?type=data&status=active&id=40981&sort=runs

https://www.openml.org/search?type=data&status=active&id=42477&sort=runs

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https://github.com/topuBhuyan/Imbalance-Aware-AutoML-for-Financial-Risk-Prediction-